

# ENPH 453 Advanced Physics Laboratory Report 1

Date of submission: February 9, 2026
Instructors: Dr. Ken Clark, Dr. Jean-Michel Nunzi, Dr. Nahee Park

Melissa Choloniuk: 20005142
Kevin Gao: 20364889
Rowan Moran: 20351280

# Abstract

Concise, clear and accurate overview of purpose, key findings and significance of the experiment.

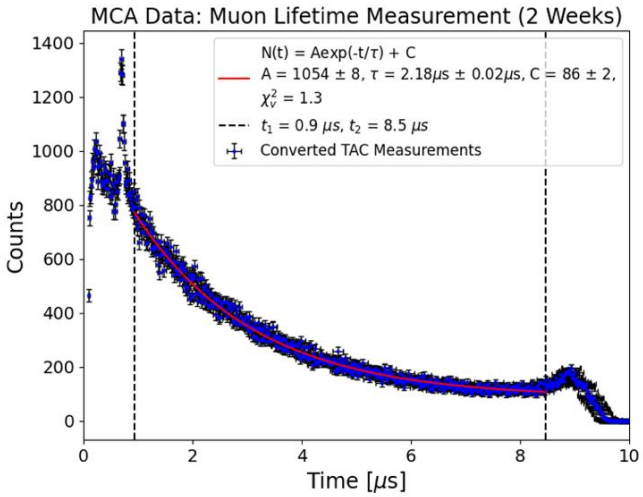
The aim of this experiment was to observe and establish the lifetime of muons by measuring the decay time of a muon as it slows down or stops inside a plastic scintillator. When a muon decays in a plastic scintillator, the muon produces a flash of light that is then detected by the photomultiplier tube (PMT). The generated flash of light is converted into electronic signals and is processed through the Nuclear Instrumental Modules (NIMS) before finally being plotted and recorded on LabVIEW. The data was plotted and fitted to find the exponential decay equation and the experimental muon lifetime was found to be 2.19703 ± 0.00004μs. This compares to the accepted theoretical value of *** and confirms the recorded value as being within an acceptable range of uncertainty.

# Executive Summary

Plot is the best representation of the work done. Summary is brief and contains the relevant information.

The purpose of this experiment was to experimentally confirm and measure the lifetime of muons. This was achieved by observing the flash that is produced when muons stop and then decay inside of a plastic scintillator. The scintillator is coupled to a photomultiplier tube which permits the decay event of the muons to be registered and recorded. Muons, being nearly identical to electrons apart from being heavier and having a shorter lifespan, decays into an electron or positron and two neutrinos. The former are responsible for the flash of light produced in the scintillator thereby enabling the PMT to convert this into an electronic pulse that is then sent to the oscilloscope to be overserved or the program to be recorded.

Need to finish



# Table of Contents

Abstract ..... i

Executive Summary ..... ii

1 Introduction ..... 1

2 Theory ..... 2

3 Apparatus and Procedure ..... 4

3.1 Apparatus ..... 4

3.2 Procedure ..... 5

4 Data Collection and Analysis ..... 7

4.1 TAC Calibration ..... 7

4.2 Muon Lifetime Measurement ..... 10

Data Collection ..... **Error! Bookmark not defined.**

5 Results and Discussion ..... 16

Results and Discussion ..... **Error! Bookmark not defined.**

Conclusion ..... 17

Appendix ..... 17

A1 TAC Calibration Data ..... 17

A2 Raw Data Files ..... 18

References ..... 18

# 1 Introduction

Muons are fundamental subatomic particles that are a part of the lepton group, possessing a heavier mass and shorter lifespan (≈2.2 μs) than the electron [1]. When high-energy charged particles from outer space (cosmic rays) collide with particles in the atmosphere, they produce pions that subsequently decay into muons [1]. These muons travel toward the Earth's surface at relativistic speeds (v ≈ 0.979c) and are detected at sea level with a typical flux of ~1$cm^{-1}s^{-1}$ [2].

The observation of muons at the Earth's surface is only possible due to relativistic time dilation. The proper lifetime $\tau$, measured in the muon's rest frame, is related to the lifetime $T$ measured in the Earth's frame with Eq. 1

$$T = \gamma\tau \tag{1}$$

where $\gamma$ is the Lorentz factor [2]. This allows muons to survive long enough to reach the ground before decaying. If special relativity were false, the time measured in the Earth frame would equal the muon's proper lifetime. In that case, the maximum distance $d$ a muon could travel would simply be the product of its speed $v$ and its lifetime, shown by Eq. 2.

$$d = v\tau = 0.98c \cdot 2.2\mu s = 646m \tag{2}$$

However, the altitude at which muons are produced is ~15km [3], far greater than this non-relativistic travel distance. The observation of a substantial muon flux at the Earth's surface therefore provides strong experimental evidence for relativistic time dilation and the validity of special relativity. In addition, measuring the muon lifetime allows for the determination of the Fermi coupling constant $G_F$, which determines the strength of the weak interaction force, and subsequently the mass of the W boson $M_W$ [4]. Therefore, measuring the muon lifetime has important implications in both relativity and particle physics.

## 2 Theory

Muons are unstable and will rapidly decay into an electron (or position) and two neutrinos. The process is mediated by the weak nuclear force, shown below in Figure 1.

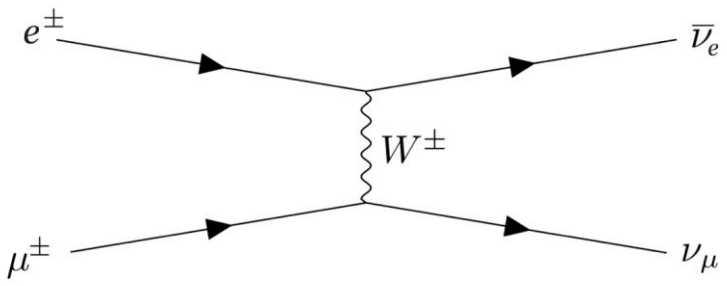



*Figure 1: Feynman diagram showing the decay process of a muon $\mu^{\pm}$, resulting in a neutrino and electron/positron pair*

When a muon slows down inside a scintillator, the deposited energy will generate a brief flash of light [5]. If it also stops, the energy released by the decay process will emit a second flash of light. These events will appear as two sharp, coincident scintillation signals separated by a short delay. By measuring the time between these events for many muons, it is expected that a graph of counts $N$ versus time $t$ will produce an exponential decay curve, given below by Eq. 3

$$N = N_0 e^{-\frac{t}{\tau}} + N_{BGND} \tag{3}$$

where $N_0$ is the number of particles at $t = 0$ and $\tau$ is the particles lifetime. By fitting this equation to the collected data, it is possible to extract the mean muon lifetime [2]. In addition, since muons are approximately 200x heavier than the electron [1], a negatively charged muon will orbit extremely close to nearby atomic nuclei. This means there is a high probability for the nuclei to capture the muon, forming a secondary decay channel and resulting in a shorter effective lifespan. This process is shown below in Figure 2.

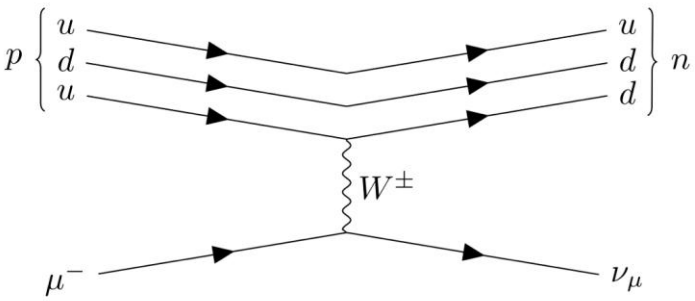



*Figure 2: Feynman diagram showing a negative muon $\mu^-$ being captured by a proton and subsequently decaying into a neutrino and neutron.*

The capture rate $\Lambda_c$ for a muon is given by the Primarkoff Formula [6] (Eq. 4)

$$\Lambda_c(A,Z) = Z_{eff}^4 X_1 \left(1 - X_2\left(\frac{A-Z}{2A}\right)\right) \tag{4}$$

where $A$ and $Z$ are the atomic mass and number respectively for the target material, $Z_{eff}$ is the effective charge the muon experiences, and $X_1 = 1.70 \cdot 10^{-4}\mu s^{-1}$, $X_2 = 3.125$ are constants obtained from fits to experimental data [6]. Using this, we can relate the capture rate to the mean negative muon lifetime $\tau_{\mu^-}$ by using Eq. 5 [7]

$$\tau_{\mu^-} = \left(\Lambda_c + \frac{Q}{\tau_\mu}\right)^{-1} \tag{5}$$

where $0 < Q \leq 1$ is the Huff factor which accounts for the muon decay rate being reduced by its atomic binding energy, and $\tau_\mu \approx 2.197\mu s$ is the lifetime of a freely decaying muon. The NE102 scintillator used in this experiment is composed of PVT plastic, which is primarily hydrogen and carbon. Since the capture rate is proportional to $Z^4$, we can assume that the contribution from hydrogen will be negligible. Using $Z_{eff} = 5.72$, $Q = 1.00$ [8], $A = 12$, and $Z = 6$ for carbon in Eq. 3 and Eq. 4:

$$\Lambda_c(12,6) = (5.72)^4 \cdot 1.70 \cdot 10^{-4}\mu s^{-1} \cdot \left(1 - 3.125 \cdot \left(\frac{12-6}{2 \cdot 12}\right)\right) = 0.0398\mu s^{-1}$$

$$\tau_{\mu^-} = \left(0.0398\mu s^{-1} + \frac{1.00}{2.197\mu s}\right)^{-1} = 2.02\mu s$$

Therefore, we expect negatively charged muons to have a mean lifetime of ~2.02μs in the plastic scintillator. With a sufficiently large dataset, it is possible to distinguish between the

contributions of positively and negatively charged muons [4]. The resulting curve will be a superposition of two decaying exponentials, given below by Eq. 6

$$N = N_0^+ e^{-\frac{t}{\tau_{\mu^+}}} + N_0^- e^{-\frac{t}{\tau_{\mu^-}}} + N_{BGND} \quad (6)$$

where $\tau_{\mu^+} = \tau_\mu \approx 2.197 \mu s$ since positive muons decay freely 100% of the time.

Apparatus and Procedure

Includes a thorough, clear, and precise description of the experimental apparatus (including a detailed and professionally formatted diagram) and the data collection procedures.

# 3 Apparatus and Procedure

## 3.1 Apparatus

A full list of apparatus is given below in Table 1

*Table 1: Names of each apparatus used in the muon lifetime measurement, along with a short summary of their function.*

| Component Name | Description |
|---|---|
| Oscilloscope | Device used to analyze and test electronic signals by reading off voltages over time |
| NE102 Plastic Scintillator | Will emit a photon when exposed to ionizing radiation like gamma rays |
| Photomultiplier Tube | Converts light into an electric signal through the photoelectric effect |
| High-Voltage Power Supply | Provides a steady supply of voltage between 0 to 3kV |
| Constant Fraction Discriminator | Outputs a signal only when the input voltage reaches above a set threshold |

| | |
|---|---|
| Rate Meter | Measures the average count rate from randomly arriving input pulses |
| Time to Amplitude Converter | Converts time between a start and a stop pulse to a linearly increasing voltage |
| Gate and Delay Generator | Outputs a signal with an adjustable delay and width for an arriving input pulse |
| Time Calibrator | Provides consistent output pulses at predefined time intervals and rates |
| Multichannel Analyzer | Converts an input analogue voltage signal into a digital output |

[PHOTO OF SETUP WIRING USED WITH LABELLING]

[DRAWN DIAGRAM SHOWING HOW EVERYTHING IS CONNECTED AND THE SIGNALS FLOW]

The plastic scintillator is coupled to a photo-multiplier tube (PMT), converting the flashes of light to an electronic signal [9]. This signal is processed by a Constant Fraction Discriminator (CFD), which suppresses background noise by rejecting pulses below a set voltage threshold [10]. The discriminator outputs serve as the START and STOP inputs to a Time-to-Amplitude Converter (TAC), which converts the time between pulses into an analog voltage [10]. The TAC successful conversion signal is sent into a gate and delay generator, with the delay adjusted so that the gate coincides with the stabilized portion of the TAC output pulse. To record data, the TAC output is digitized by a Multi-Channel Analyzer (MCA) [10], with the triggering input fed from the gate and delay generator. A LabVIEW program is used to record data from the MCA output. [NOT SURE IF THIS SECTION IS NECESSARY ANYMORE AFTER THE SUMMARY, ALSO PROBABLY NEED TO EXPLAIN INDIVIDUAL APPARATUS MORE]

## 3.2 Procedure

The following procedure outlines the necessary steps for repeating the experiment. Ensure that all device calibration settings are recorded and held constant after each step. [MAYBE NOT NEEDED]

1. Connect the input of the photomultiplier tube to the high-voltage power supply and the output to the oscilloscope. Turn on the voltage supply and gradually increase the voltage until a significant number of pulses are observed on the oscilloscope.

2. Plug the output of the photomultiplier tube into the input of the constant fraction discriminator (CFD) and set it to the scintillation mode. Plug the positive output of the CFD into the rate meter and slowly increase the threshold on the CFD until the count rate plateaus. This occurs when the count rate does not noticeably decrease under a small increase in the threshold.
3. Set the range of the time to amplitude converter (TAC) well above the expected lifetime of a muon (~10,000ns). Connect the START and STOP outputs from the time calibrator into the corresponding inputs on the TAC. Turn on the time calibrator and check the TAC output signal using an oscilloscope to verify its functionality. The signal should look like a square wave with a sharp, linear increase at the beginning.
4. Connect the successful TAC conversion signal from the back of the device to the gate and delay input. Connect the TAC output and gate and delay generator output into the oscilloscope. Set the trigger mode on the oscilloscope to normal and rising edge, and have it trigger on the gate and delay generator output. Adjust the gate delay until the successful conversion signal is overlapping with the flat region of the TAC output and ensure that this is well after the initial plateau point. Adjust the gate width so that the successful conversion signal is wide but not exceeding the length of the TAC output signal.
5. Connect the output from the TAC into the MCA input port and connect the gate and delay output into the MCA trigger input. On the LabVIEW program, set the bin count to 1000 and voltage range between 0 to 2V. Record data for a minimum of five different time calibrator settings where the rate and interval are set equal to each other. Run each configuration until a well-defined Gaussian peak is formed. Export the data and perform a linear fit, plotting time (interval) as a function of peak voltage value. This will give you a conversion factor between voltage and time.
6. Connect the two negative CFD outputs into the START and STOP inputs of the TAC and repeat the gate and delay calibration process from step 4.
7. Connect the TAC output to MCA input, and the gate and delay output to the MCA trigger input. On the LabView program, set the bin count to 1000 and voltage range between 0 to 10V and begin recording data. If everything is setup correctly, the data should slowly form an exponential decay curve over the span of a few hours. Use this and the voltage-to-time conversion factor to extract the mean muon lifetime.

# 4 Data Collection and Analysis

The raw data files for all measurements are provided in Appendix .

## 4.1 TAC Calibration

The TAC calibration measurements (steps 4 and 5) were performed with the following instrument settings (Table 2).

*Table 2: NIM settings for TAC calibration measurements, performed using a time calibrator.*

| Instrument | Setting |
|---|---|
| Time-to-Amplitude Converter | 10,000ns range |
| Time Calibrator | Time intervals between 0.08μs to 2.56μs |
| | Range set equal to interval |
| Gate and Delay Generator | 780ns delay |
| | 800ns width |

The LabVIEW program was run at a 250kHz frequency, 1000 bins, and a voltage range between 0 to 2V. Each time interval was run until the peak value reached ~500 counts. The collected data is plotted below in Figure 3 and tabulated in Appendix . The y-error in each bin $\sigma_N$ is given by assuming Poisson statistics (Eq. 7)

$$\sigma_N = \sqrt{N} \tag{7}$$

where $N$ is the number of counts in a bin.

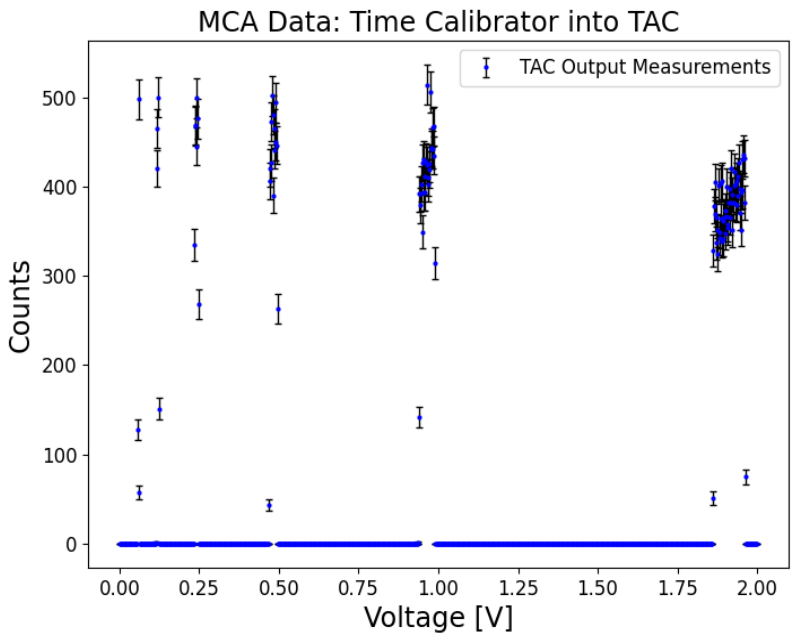



*Figure 3: Data obtained from TAC calibration measurements using a time calibrator set at six different time intervals (0.08μs to 2.56μs) and a gate delayed trigger. Plotted are the counts (y-axis) against voltage (x-axis, V) across 1000 bins and a voltage range between 0 to 2V. Error bars in the y-direction are plotted assuming Poisson statistics ($\sigma = \sqrt{N}$)*

Using the Gaussian analyzer in the LabVIEW program, the full-width half max (FWHM), peak voltage value, and total counts of each Gaussian curve was also recorded. Since the TAC output is linear in voltage $V$ and time $t$, a conversion factor between the two quantities is obtained by fitting a straight line which follows Eq. 8

$$V = mt + b \tag{8}$$

where $m$ is the slope and $b$ is the intercept. Each measurement produced a normal distribution, so the error in the peak voltage $\sigma_V$ is given by the mean error of a Gaussian, shown below by Eq. 9

$$\sigma_V = \frac{\sigma}{\sqrt{N}} \tag{9}$$

where $N$ is the total number of counts (area) in each Gaussian curve, and $\sigma$ is the standard deviation given by Eq. 10.

$$\sigma = \frac{FWHM}{\sqrt{2ln2}} \tag{10}$$

The error in the time interval $\sigma_t$ was set equal to ±0.005%, quoted directly from the time calibrator specifications sheet [11]. A linear regression using the total least squares method was performed, and the results are shown below in Figure 4 and Table 3. The last Gaussian curve, which corresponds to a time interval of 2.56μs, was removed from analysis due to an ill-defined peak.

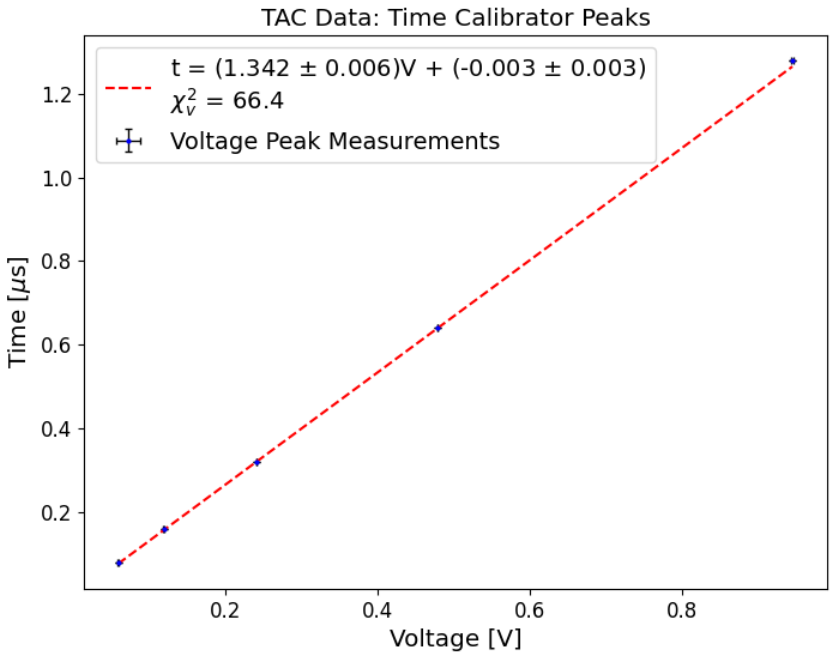



*Figure 4: TAC calibration data plotting the peak voltages (V) corresponding to each time calibrator interval (μs). A straight was fitted following $V = mt + b$ using the total least squares method, shown by the dashed red line. A reduced chi squared value is also provided. Errors in the x and y directions are plotted but are too small to be discernable.*

*Table 3: Results of linear regression for the TAC calibration.*

| Parameter | Value | Error (±) |
|---|---|---|
| $\boldsymbol{m}$ | 1.342 | 0.006 |
| $\boldsymbol{b}$ | -0.003 | 0.003 |
| $\boldsymbol{\chi_v^2}$ | 66.4 | N/A |

The high reduced chi squared value of $\chi_v^2 = 66.4$ is caused by the small error in both voltage and time, as well as the small number of degrees of freedom (5) from only using the peak voltages instead of the full dataset. The waveforms outputted from the time calibrator have to travel to the TAC and be processed, likely leading to a higher time interval error than expected. Alongside this, it is evident (Figure 3) that for smaller voltages the bin widths are too short to specify a well defined FWHM. Despite this, the data shows a linear trend as expected, so the equation was used in later analysis.

## 4.2 Mean Muon Lifetime Measurement

A 2 weeklong data collection of muon decays were performed with the following instrument settings (Table 4).

*Table 4: NIM settings which were held constant during the muon lifetime measurement.*

| Instrument | Setting |
|---|---|
| High-Voltage Power Supply | 2400V |
| Time-to-Amplitude Converter | 10,000ns range |
| Constant Fraction Discriminator | 900mV threshold |
| Gate and Delay Generator | 780ns delay |
| | 800ns and 400ns width |

Two separate, weeklong runs were performed with an 800ns and 400ns gate width respectively. The LabVIEW program was run with a 250kHz frequency, 1000 bins, and a voltage range between 0 to 8V. The maximum voltage was chosen after a preliminary 2 daylong run showed that the TAC output ended around ~7V. The combined data is plotted below in Figure 4Figure 5, and provided in Appendix . Poisson statistics were assumed for the y-error (Eq. 8).

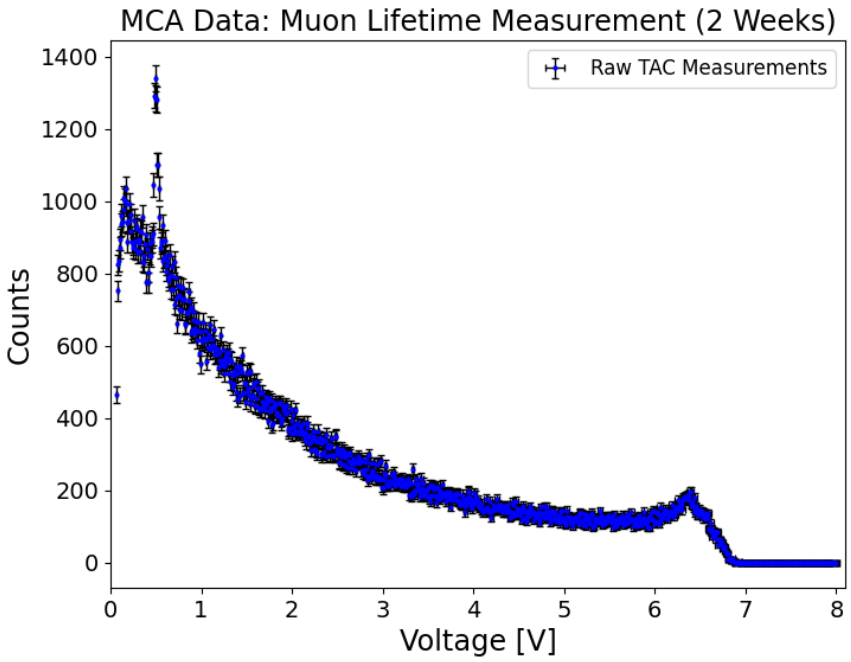



*Figure 5: Data obtained from an MCA for the muon lifetime measurement, where a scintillator and PMT setup is used. The measurement is read from a TAC, which converts the decay time into a voltage signal. Plotted are the counts (y-axis) versus voltage (x-axis, V) for a 2 weeklong measurement, with 1000 bins and a voltage range between 0 to 8V. Error bars in the y-direction are plotted assuming Poisson statistics ($\sigma = \sqrt{N}$)*

The voltages were converted into time using Eq. 8, and error was propagated using the derivative method. Weighted curve fitting was performed using SciPy's optimization package and an exponential decay formula, given below by Eq. 11

$$N(t) = Ae^{-\frac{t}{\tau}} + C \tag{11}$$

where $A$, $\tau$, and $C$ are fitting parameters. Erroneous regions, including the sharp peak at ~0.8V and the smooth bump at ~6.5V, were excluded from the fit. Justifications for this are discussed in Section 5. The resulting fit is shown below in Figure 6.

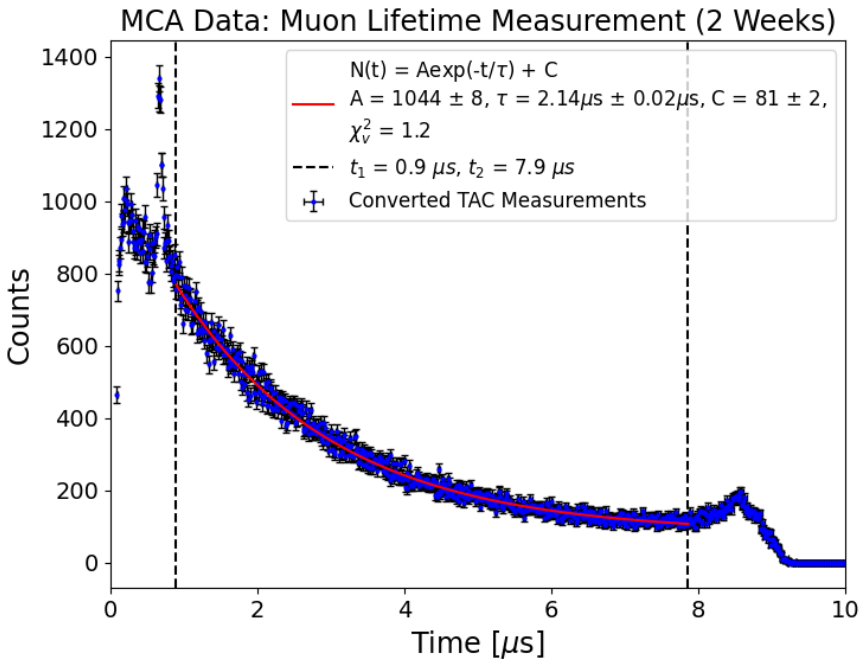



*Figure 6: Data obtained from an MCA for the muon lifetime measurement, where a scintillator and PMT setup is used. The voltage measurements from the TAC were converted into time using Eq. 8. Plotted are the counts (y-axis) versus time (x-axis, μs) for a 2 weeklong measurement, with 1000 bins and a voltage range between 0 to 8V. An exponential decay curve (Eq. 11) was fitted between 0.9μs to 7.9μs and overlayed on top of the data. Error bars in the x and y directions are also included.*

The resulting fitting parameters are shown below in Table 5.

*Table 5: Results of curve fitting for the muon lifetime measurement, following an exponential decay formula (Eq. 11).*

| Parameter | Value | Error (±) |
|---|---|---|
| $A$ | 1.342 | 0.006 |
| $\tau$ | 2.14 μs | 0.02 μs |
| $C$ | 81 | 2 |
| $\chi^2_v$ | 1.2 | N/A |

**Commented [RM1]:** No way this number is right.

The standardized residuals $r$ were also calculated, given by Eq. 12 below

$$r = \frac{t - \hat{t}}{\sigma_V} \tag{12}$$

where $t$ are the measured times, and $\hat{t}$ are the expected times given by the fitted curve. This is plotted below in Figure 7.

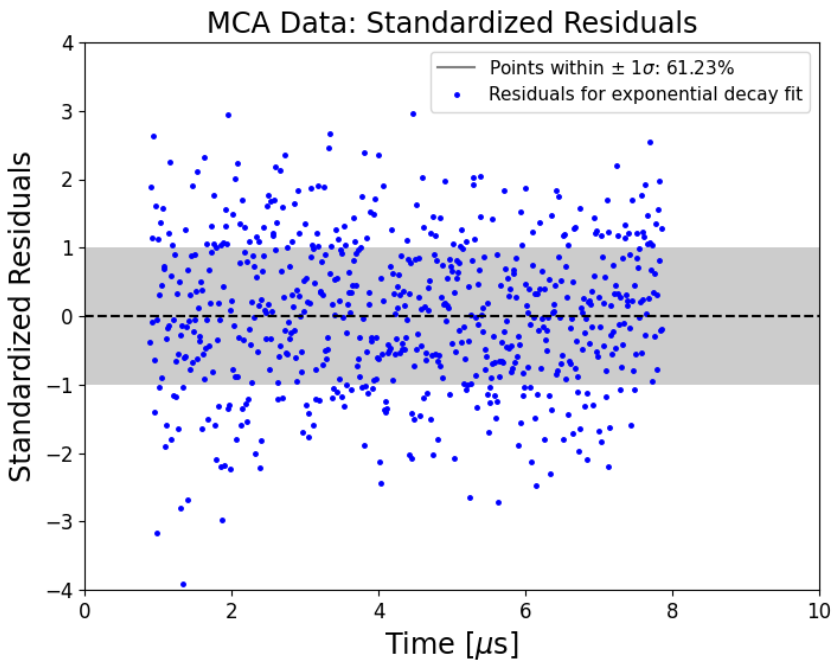



*Figure 7: Standardized residuals for the muon lifetime measurement, where a scintillator and PMT setup is used. The voltage measurements from the TAC were converted into time using Eq. 8. Plotted are the residuals (y-axis) versus time (μs) in the region of interest (0.9μs to 7.9μs). The difference was taken between the fitted exponential decay curve and the data and divided by the standard deviations of each point.*

Using the standardized residuals, 61.23%, 99.69%, and 100% of the data lie within 1σ, 3σ, and 5σ, respectively. This indicates the exponential decay model is a good fit to the data and is also statistically significant, consistent with the standard 5σ criterion in particle physics [12].

## 4.2 Positive and Negative Muon Lifetime Measurement

Since the detected muon decays consist of both positively charge muons ($\mu^+$), which decay freely, and negative charge muons ($\mu^-$), which may undergo nuclear capture in addition to decay, the total decay spectrum is expected to fit a double-exponential decay model.

Thus, the collected data from the 2 weeklong detection of muon decays plotted above in Figure 4Figure 5 was fit to a model of the form:

$$N(t) = Ae^{-\frac{t}{\tau_1}} + Be^{-\frac{t}{\tau_2}} + C \tag{13}$$

where $\tau_1$ and $\tau_2$ are the effective lifetimes of the two muon populations, A and B are their respective amplitudes, and C accounts for a constant background.

As in the single-exponential analysis, the data was truncated before 0.9μs and after 7.9μs to exclude non-physical regions discussed in Section 5. The resulting fit is shown below in Figure *.

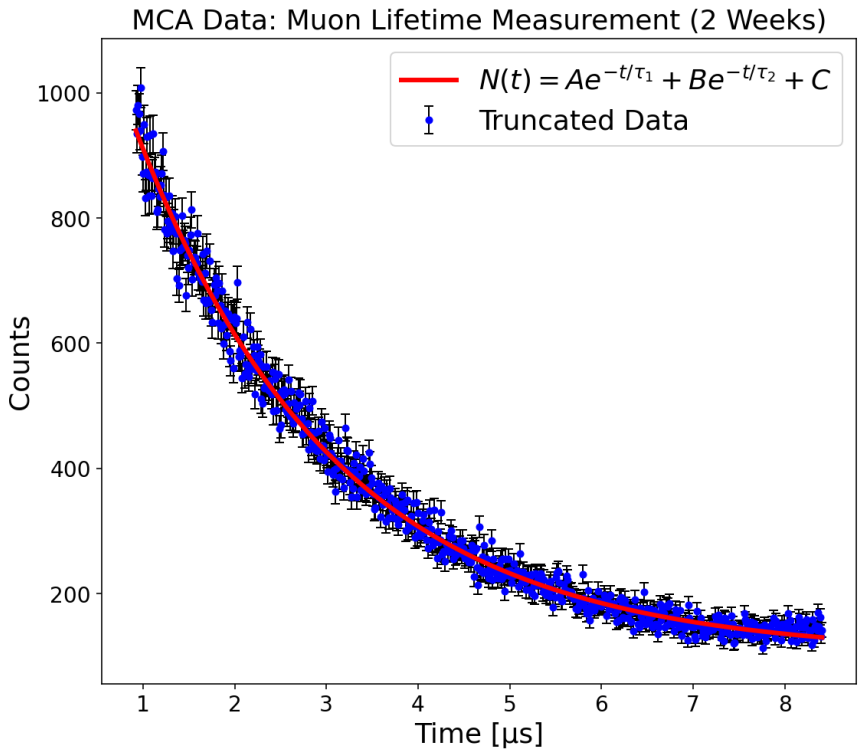



*Figure 8: 2 weeklong data obtained from an MCA for the muon lifetime measurement, where a scintillator and PMT setup is used. A double exponential decay curve (Eq. 13) was fitted between 0.9μs to 7.9μs and overlayed on top of the data. Error bars in the x and y directions are also included.*

The resulting fitting parameters for Eq. 13 are shown below in Table *.

*Figure 9: Results of curve fitting for the muon lifetime measurement, following a double exponential decay formula (Eq. 13).*

| Parameter | Value | Error (±) |
|---|---|---|
| $\boldsymbol{A}$ | 500 | $2 \times 10^8$ |
| $\boldsymbol{B}$ | 800 | $2 \times 10^8$ |
| $\boldsymbol{\tau_1}$ | 2.18 μs | 500 μs |
| $\boldsymbol{\tau_2}$ | 2.18 μs | 300 μs |
| $\boldsymbol{C}$ | 103 | 6 |
| $\boldsymbol{\chi_v^2}$ | 1.3 | N/A |

The fitted decay constants were found to be approximately ***. The reduced chi-squared value of 1.3 indicates that the model provides an acceptable overall fit for the data. However, the fitted uncertainties on both the amplitudes, $A$ and $B$, and decay constants, $\tau_1$ and $\tau_2$, were found to be extremely large, several orders of magnitude higher than their respective value. This indicates that the parameters are poorly constrained by the dataset, despite the acceptable goodness-of-fit statistic.

To investigate the origin of the large parameter uncertainties, the reduced

Data analysis is meticulous, complete, and accurate. Good fit to data (if applicable). Thorough discussion of uncertainties

*ROWAN: Plot of double exponential fit and goodnesses of fit (reduced chi squared plots)

Table: Summary of results for fixing different fitting parameters for double exp.

| Trial | Fixed Parameter | Curve Fitting Parameter | Value | Uncertainty (±) |
|---|---|---|---|---|
| 1 | $A = B$ | $\boldsymbol{A}$ | 640 | 10 |
| | | $\boldsymbol{\tau_1}$ | 2.18 μs | 6000 μs |
| | | $\boldsymbol{\tau_2}$ | 2.18 μs | 6000 μs |
| | | $\boldsymbol{C}$ | 103 | 6 |
| 2 | $A=B$ | $\boldsymbol{A}$ | 641 | 4 |
| | | $\boldsymbol{\tau_2}$ | 2.34 μs | 0.04 μs |

| | | | |
|---|---|---|---|
| $\tau_1 = 2.02\ \mu s$ | $C$ | 102 | 2 |

## 5 Results and Discussion

* ROWAN : Discussion of systematic error at start and end of data. Discussion of goodness of fit and tau values for both single and double exponential. Discussion of muon flux through scintillator and how it compares with ratemeter count at correct CFD threshold. Discussion of how long program would need to run to determine tau1 and tau2 with higher accuracy.

Includes an insightful comparison of experimental results to theoretical predictions and a description of the experimental limitations. Discussion accounts for any inconsistent or confounding findings. Limitations of theoretical predictions are addressed.

Questions:

Theoretical Considerations T.1) Estimate the rate of muons passing through the plastic scintillator by measuring the size of the detector and assuming a muon flux ≈ 1 cm−2min−1 at Stirling Hall. T.2) What is the uncertainty on the final resulting count bins? T.3) How will you take the random coincidences into account (if you feel you need to do so)? Experimental Considerations E.1) Make sure you understand how each component works and what is connected to what. The output pulse shapes of each electronics unit should be examined and recorded using an oscilloscope to gain an understanding of the operation of the circuit. E.2) At what level should the discriminator be set to remove as much background as possible? Can you test this level? E.3) Calibrate the system using the provided calibration device. How much data will you need? E.4) Once you have collected some data, determine how long the experiment should run to achieve the result you want.

# 6 Conclusion

* ROWAN :

Conclusion is well
- supported by the
experimental data.
Implications are discussed
within a theoretical
framework.

Overall

The report is exceptionally
well structured and
organized, and it is
professionally formatted.
Nearly no technical writing
or citation errors.

# Appendix

## A1 TAC Calibration Data

*Table 6: Data used in TAC calibration analysis derived from raw data. Time intervals were set on the time calibrator.*

| Time Interval [$\mu s$] | Peak Voltage [V] | Counts | FWHM [V] |
|---|---|---|---|
| 0.080000 ± 0.000004 | 0.0590 ± 0.0006 | 498 | 0.034 |
| 0.160000 ± 0.000008 | 0.1190 ± 0.0005 | 465 | 0.024 |

| | | | |
|---|---|---|---|
| 0.32000 ± 0.00002 | 0.2410 ± 0.0002 | 499 | 0.010 |
| 0.64000 ± 0.00003 | 0.47900 ± 0.00004 | 502 | 0.002 |
| 1.28000 ± 0.00006 | 0.945 ± 0.001 | 514 | 0.054 |
| 2.5600 ± 0.0001 | 1.8690 ± 0.0009 | 369 | 0.042 |

## A2 Raw Data Files



Raw_Data_E1.zip

## References


[1] Y. Kuno and Y. Okada, "Muon decay and physics beyond the standard model," *Rev. Mod. Phys.*, vol. 73, no. 1, pp. 151–202, Jan. 2001, doi: 10.1103/RevModPhys.73.151.

[2] U. Latif, M. S. Anwar, and I. Younas, "Measuring the lifetime of cosmic ray muons".

[3] "Muon - an overview | ScienceDirect Topics." Accessed: Feb. 07, 2026. [Online]. Available: https://www.sciencedirect.com/topics/chemistry/muon

[4] "muon.pdf." Accessed: Feb. 03, 2026. [Online]. Available: https://www.physics.utoronto.ca/apl/muon/muon.pdf

[5] "Muon Physics." Accessed: Jan. 14, 2026. [Online]. Available: https://web.njit.edu/~sirenko/Phys450/MU.pdf

[6] T. Suzuki, D. F. Measday, and J. P. Roalsvig, "Total nuclear capture rates for negative muons," *Phys. Rev. C*, vol. 35, no. 6, pp. 2212–2224, Jun. 1987, doi: 10.1103/PhysRevC.35.2212.

[7] R. Mizuno *et al.*, “Lifetime measurement of the muonic atoms of enriched Si isotopes,” *Phys. Rev. C*, vol. 112, no. 2, p. 024307, Aug. 2025, doi: 10.1103/vl7z-rzp8.
[8] K. W. Ford and J. G. Wills, “Calculated properties of μ-mesonic atoms,” *Nuclear Physics*, vol. 35, pp. 295–302, Jun. 1962, doi: 10.1016/0029-5582(62)90113-X.
[9] M. Li *et al.*, “Performance of compact plastic scintillator strips with wavelength shifting fibers using a photomultiplier tube or silicon photomultiplier readout,” *NUCL SCI TECH*, vol. 34, no. 2, p. 31, Mar. 2023, doi: 10.1007/s41365-023-01175-6.
[10] D. Bailey and D. Paul, “ADVANCED UNDERGRADUATE LABORATORY EXPERIMENT 35 MUON LIFETIME,” Nov. 1996.
[11] “462 Time Calibrator | Electronics | AMETEK ORTEC.” Accessed: Feb. 07, 2026. [Online]. Available: https://www.ortec-online.com/products/electronic-instruments/time-to-amplitude-converters/462
[12] “Why do physicists mention ‘five sigma’ in their results? | CERN.” Accessed: Feb. 07, 2026. [Online]. Available: https://home.cern/resources/faqs/five-sigma